

Prashant Rahul

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Work experience

TinkerBloX, Coimbatore *(Remote)*

Aug 2026 – present

Research and Development System Engineer

- Developing a low-level container runtime in Rust, using Linux namespaces, cgroups, and other kernel primitives.

Cypherock, Gurugram *(Remote)*

Aug 2025 – Jan 2026

Software Engineer Intern

- Implemented the Canton Network within STM32 embedded firmware for the X1 crypto hardware wallet, including writing real time encoders and decoders, hashing and signing Canton transactions.
- Contributed to the Electron frontend application, developing features using Typescript.

Open Source & Programs

Linux Kernel Summer 2026 - The Linux Foundation

- Improved the Octeon staging Ethernet driver by replacing panic() calls with proper error propagation and resource unwinding on probe failures.
- Updated the TI ADC081C and ADC084S021 drivers to use dev_err_probe(), simplifying probe error handling and supporting deferred probing.

RTEMS - RTOS for Multiprocessor Systems

- Enabled IMFS testing on BSPs with tight memory constraints, like the sparc/erc32 bsp
- Implemented POSIX compliant ENAMETOOLONG handling in cpukit/libio by validating path components against NAME_MAX during filesystem path evaluation

The Julia Programming Language

- Improved performance in show() for UnitRange append! for AbstractVector
- Quality of life improvements when displaying empty ranges

Education

Bachelor's Degree in Computer Science

2027

CV Raman Global University

Projects

KaworuOS *(aarch64, kernel, unix-like)*

A WIP hobby arm64 Unix-like operating system with preemptive multitasking and multiprocessor support.

Skills

Programming languages: Rust, C/C++, Python, Go, LLVM IR, QBE IR, Typescript, Javascript

Technologies & Tools: GNU/Linux, Nix/NixOS, Git, CMake, Make, Meson, Vi/Vim/Neovim, POSIX Shells, NodeJS

Concepts & Practices: Embedded Systems, RTOS, Device drivers, Firmware development, Operating systems, Cryptography Fundamentals, ARM ISA, Version Control, CI/CD